Coding & Solution

Date	27June 2026
Team ID	
Project Name	AComprehensive Measure of Well-Being
Maximum Marks	5Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	***Details***
Repository Link / URL	https://github.com/Tejagunupuru18/HDIProject.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask (Backend), Scikit-Learn (Machine Learning), Pandas (Data Processing)
Key Features Implemented	• TrainedLinearRegressionmodel(hdi_model.pkl)tocalculateHDI scores. • Flask REST API endpoint to handle backend data routing. • Responsive web dashboard for inputting 4 key socio-economic metrics. • Live cloud deployment on Render.
Pending / Incomplete Features	• Exporting prediction results directly to PDF. • Advanced interactive charts for historical data comparison.
Setup / Run Instructions	1. Clone the repository from GitHub. 2. Navigate to the project folder and run: pipinstall-r requirements.txt 3. Start the local development server by running:

Code Quality Checklist

S.No	***Criteria***	***Status (Yes / No)***
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The solution successfully integrates a Scikit-Learn machine learning model with a Flask web backend to deliver real-time Human Development Index predictions. By decoupling the trained model (hdi_model.pkl) from the web routing logic, the architecture remains highly modular. The final web application handles data normalization safely and is fully deployed and accessible to end-users via the Render cloud platform.